

Abstract

We measured the local acceleration due to gravity, g , by timing the small-angle oscillations of a simple pendulum of variable length. Using a linear fit of period squared against length, we obtained $g = 9.79 \text{ m s}^{-2}$, within 0.2 % of the accepted value of 9.81 m s^{-2} . Sources of systematic error, including air resistance and finite string mass, are discussed in Section 4.

The slope, once fit to data, yields g directly.¹

¹This linearization avoids the bias introduced by fitting T against $\sqrt{L}$ directly, since ordinary least squares assumes additive, not multiplicative, noise.